Experiment 3 — Stronger CNN Backbones on Colab

This notebook compares ImageNet-pretrained ResNet50 and ConvNeXt-Tiny under the same split, preprocessing and checkpoint criterion. Run one model at a time if Colab runtime is limited.

1. Configuration

```
Mode   : Local
Device : NVIDIA RTX PRO 2000 Blackwell Generation Laptop GPU
```

2. Stage images and verify the shared split

	rows	classes
train	62312	27
validation	16984	27

3. Use identical data transformations for a fair comparison

4. Build each model and expose its final block

5. Train the head, then the final feature block

```
===== resnet50 =====
```

	model	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	resnet50	1	head	1	1.5646	1.3831	0.5901	0.5838	0.5385

	model	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	resnet50	2	last_block	1	1.1870	1.1311	0.6583	0.6548	0.6228

	model	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	resnet50	3	last_block	2	0.8600	1.1223	0.6705	0.6692	0.6372

	model	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	resnet50	4	last_block	3	0.6110	1.1799	0.6723	0.6707	0.6415

	validation_loss	accuracy	weighted_f1	macro_f1
resnet50	1.1799	0.6723	0.6707	0.6415

	prdtypecode	precision	recall	f1	support
8	1281	0.393	0.292	0.335	414
24	2585	0.437	0.443	0.440	499
7	1280	0.467	0.436	0.451	974
22	2582	0.455	0.512	0.482	518
6	1180	0.553	0.444	0.493	153
11	1302	0.579	0.454	0.509	498
2	50	0.480	0.548	0.512	336
17	2220	0.708	0.455	0.554	165
16	2060	0.560	0.568	0.564	999
12	1320	0.593	0.549	0.571	648
0	10	0.506	0.713	0.592	623
13	1560	0.653	0.579	0.614	1015
1	40	0.681	0.612	0.644	502
20	2462	0.737	0.602	0.663	284
15	1940	0.608	0.733	0.665	161
4	1140	0.702	0.680	0.691	534
19	2403	0.689	0.715	0.702	955
21	2522	0.710	0.727	0.719	998
3	60	0.691	0.753	0.720	166
9	1300	0.666	0.839	0.743	1009
26	2905	0.780	0.713	0.745	174
18	2280	0.775	0.749	0.762	952
10	1301	0.856	0.702	0.771	161
25	2705	0.820	0.761	0.789	552
14	1920	0.773	0.849	0.809	861
23	2583	0.858	0.837	0.847	2042
5	1160	0.955	0.919	0.937	791

```
===== convnext_tiny =====
```

	model	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	convnext_tiny	1	head	1	1.5254	1.3740	0.5894	0.5816	0.5427

	model	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	convnext_tiny	2	last_block	1	1.2034	1.1308	0.6572	0.6545	0.6211

	model	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	convnext_tiny	3	last_block	2	0.8486	1.1198	0.6686	0.6657	0.6327

	model	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	convnext_tiny	4	last_block	3	0.5391	1.1671	0.6766	0.6739	0.6458

	validation_loss	accuracy	weighted_f1	macro_f1
convnext_tiny	1.1671	0.6766	0.6739	0.6458

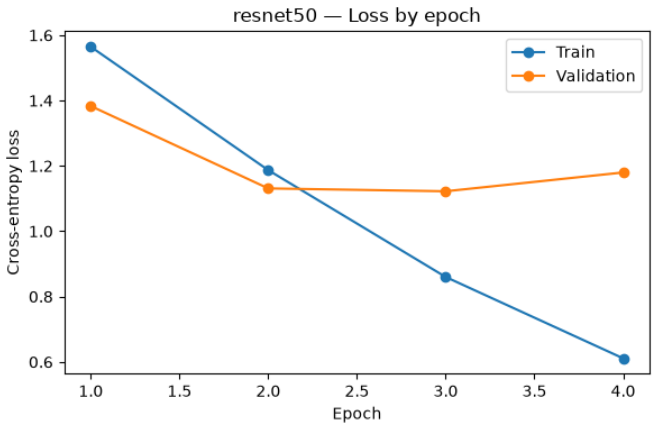
	prdtypecode	precision	recall	f1	support
8	1281	0.411	0.350	0.378	414
24	2585	0.480	0.439	0.459	499
7	1280	0.479	0.451	0.464	974
22	2582	0.504	0.459	0.481	518
6	1180	0.606	0.431	0.504	153
11	1302	0.506	0.506	0.506	498
17	2220	0.580	0.503	0.539	165
0	10	0.584	0.549	0.566	623
16	2060	0.558	0.576	0.567	999
2	50	0.608	0.536	0.570	336
12	1320	0.570	0.582	0.576	648
1	40	0.557	0.625	0.589	502
13	1560	0.645	0.630	0.637	1015
20	2462	0.758	0.563	0.646	284
26	2905	0.732	0.580	0.647	174
19	2403	0.641	0.718	0.678	955
4	1140	0.714	0.674	0.694	534
21	2522	0.737	0.735	0.736	998
3	60	0.729	0.747	0.738	166
18	2280	0.743	0.748	0.746	952
9	1300	0.689	0.812	0.746	1009
15	1940	0.771	0.752	0.761	161
25	2705	0.806	0.797	0.801	552
10	1301	0.877	0.752	0.809	161
14	1920	0.784	0.859	0.820	861
23	2583	0.845	0.871	0.858	2042
5	1160	0.926	0.919	0.923	791

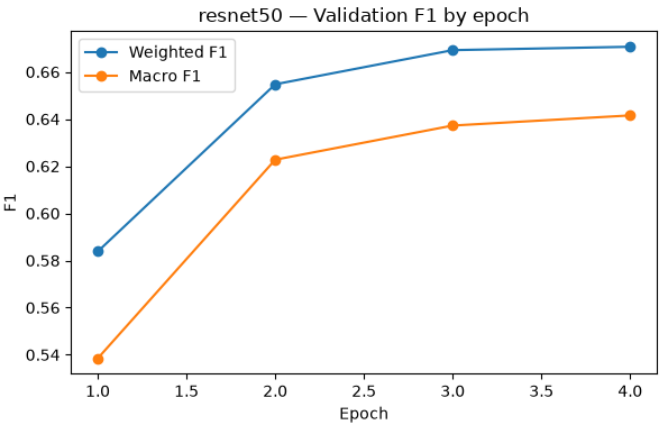
	model	validation_loss	accuracy	weighted_f1	macro_f1
1	convnext_tiny	1.1671	0.6766	0.6739	0.6458
0	resnet50	1.1799	0.6723	0.6707	0.6415

6. Training curves

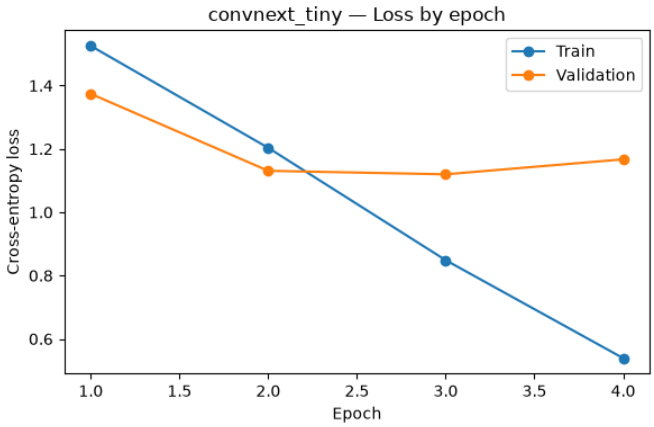
Use validation loss and macro F1 to assess generalisation for each backbone. A falling training loss combined with a rising validation loss indicates overfitting.

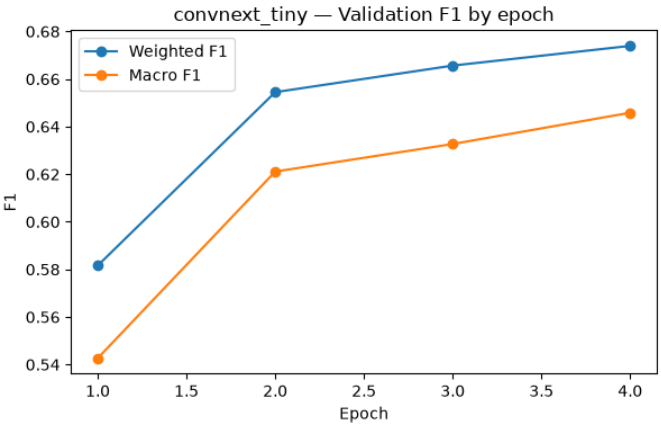
	model	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
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1	resnet50	2	last_block	1	1.1870	1.1311	0.6583	0.6548	0.6228
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3	resnet50	4	last_block	3	0.6110	1.1799	0.6723	0.6707	0.6415





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1	convnext_tiny	2	last_block	1	1.2034	1.1308	0.6572	0.6545	0.6211
2	convnext_tiny	3	last_block	2	0.8486	1.1198	0.6686	0.6657	0.6327
3	convnext_tiny	4	last_block	3	0.5391	1.1671	0.6766	0.6739	0.6458





Interpretation

Select the backbone using macro F1 first and weighted F1 second. Because both candidates are evaluated on the same 16,984 rows, the scores and per-class changes are directly comparable.